

[illegible]

1. RVT #1 — renal vein thrombosis

WHAT YOU SEE

Gargoyle: RVT #1; PE, DVT, ALB ↓ → risk

WHAT IT ENCODES

Membranous nephropathy is the nephrotic disease MOST associated with hypercoagulability and renal vein thrombosis, plus DVT/PE. Mechanism: urinary loss of antithrombin III and other anticoagulant proteins. Flank pain + hematuria + worsening proteinuria in a membranous patient = think RVT.

BOARD TRAP

RVT/hypercoagulability is the membranous hallmark — not MCD or FSGS. Low albumin amplifies the clotting risk.

2. Primary = subepithelial only

WHAT YOU SEE

Banner: PRIMARY = SUBEPITHELIAL ONLY

WHAT IT ENCODES

In primary (PLA2R) membranous, immune complexes are exclusively subepithelial — no mesangial or subendothelial deposits. IF shows granular IgG (IgG4-dominant) and C3 along the GBM. EM: subepithelial deposits with GBM spikes between them.

BOARD TRAP

Subepithelial-ONLY = primary; extra-membranous (mesangial/subendothelial) deposits point to SECONDARY (lupus, infection, malignancy).

3. Secondary → extra sites + IgG1-3

WHAT YOU SEE

Label: SECONDARY = EXTRA SITES + IgG1-3

WHAT IT ENCODES

Secondary membranous (lupus, HBV, syphilis, solid tumors, NSAIDs/gold) shows deposits in extra locations (mesangial/subendothelial), full-house IF in lupus, and IgG1/IgG2/IgG3 subclasses rather than primary's IgG4. PLA2R usually negative.

BOARD TRAP

IgG1-3 + extra deposit sites + full-house = SECONDARY/lupus; IgG4 + subepithelial-only + anti-PLA2R = primary.

4. Anti-PLA2R 70–80% / no biopsy

WHAT YOU SEE

Stained-glass figure: PLA2R 70-80%; ANTI-PLA2R → NO BIOPSY

WHAT IT ENCODES

Anti-PLA2R antibodies are positive in ~70–80% of primary membranous and are diagnostic — a positive titer in a nephrotic patient may obviate biopsy. THSD7A accounts for a minority. Titer tracks disease activity and predicts response/relapse.

BOARD TRAP

Anti-PLA2R = primary membranous biomarker; it is NOT seen in MCD, FSGS, or lupus membranous.

5. Silver stain: spike-and-dome

WHAT YOU SEE

SILVER STAIN, SPIKE + DOME diagram

WHAT IT ENCODES

Jones methenamine silver stain shows GBM 'spikes' projecting between subepithelial deposits, creating the spike-and-dome appearance. EM confirms electron-dense subepithelial deposits.

BOARD TRAP

Spike-and-dome on silver = membranous; tram-tracks (double contour) on silver = MPGN. Don't confuse the two silver patterns.

6. IgG + C3 granular IF

WHAT YOU SEE

IgG ONLY / IgG + C3 labels

WHAT IT ENCODES

Immunofluorescence shows granular (lumpy-bumpy) IgG and C3 along the capillary loops — classic immune-complex pattern. Primary disease is IgG4-dominant.

BOARD TRAP

Granular IgG/C3 along GBM = immune-complex membranous; linear IgG = anti-GBM (Goodpasture), a totally different disease.

7. Rituximab (anti-PLA2R driven)

WHAT YOU SEE

Monk figure: RITUXIMAB

WHAT IT ENCODES

Rituximab (B-cell/anti-CD20) is first-line immunosuppression for high-risk primary membranous, depleting PLA2R antibody-producing B cells. Anti-PLA2R titer falls before proteinuria improves and is used to monitor response.

BOARD TRAP

Rituximab targets the B cells making anti-PLA2R; it's preferred over older Ponticelli cyclophosphamide+steroid in many high-risk patients.

8. Ponticelli: steroids + cyclophosphamide

WHAT YOU SEE

PONTICELLI: STEROIDS + CYCLOPHOSPHAMIDE

WHAT IT ENCODES

The Ponticelli regimen alternates monthly corticosteroids with cyclophosphamide for high-risk primary membranous. CNIs (cyclosporine/tacrolimus) are an alternative. Reserved for high-risk (heavy proteinuria, rising anti-PLA2R, declining GFR).

BOARD TRAP

Immunosuppression is for HIGH-RISK membranous only — low-risk patients get conservative RAAS therapy first.

9. Risk-stratified: very high / high / mod / low

WHAT YOU SEE

Color tower: VERY HIGH-URGENT IS, HIGH-START IS, MOD-REASSESS, LOW-WATCH

WHAT IT ENCODES

Treatment is risk-based: low risk (preserved GFR, proteinuria <3.5 g, stable anti-PLA2R) → supportive RAAS and watch (one-third remit spontaneously). High/very-high risk → start immunosuppression.

BOARD TRAP

Don't immunosuppress every membranous — ~1/3 remit spontaneously, so low-risk patients are watched on RAAS.

10. C3/C4 normal unless lupus

WHAT YOU SEE

Gauges: C3/C4 normal; UNLESS LUPUS (red)

WHAT IT ENCODES

Primary membranous has NORMAL complement. Low C3/C4 in a 'membranous' picture should redirect you to a secondary cause — especially lupus membranous (ISN/RPS class V), which is full-house with low complement.

BOARD TRAP

Low complement + membranous pattern = lupus (secondary), not idiopathic PLA2R membranous.

11. >3.5 g nonselective proteinuria

WHAT YOU SEE

Lower-left: >3.5g NONSELECTIVE

WHAT IT ENCODES

Membranous produces heavy, NONselective proteinuria (albumin + larger proteins) and bland urinary sediment, often with edema and hyperlipidemia. Contrasts with MCD's highly selective albuminuria.

BOARD TRAP

Selective (albumin-predominant) proteinuria = MCD; nonselective heavy proteinuria = membranous/FSGS.

12. White adults — most common primary

WHAT YOU SEE

Right banner: WHITE ADULTS #1

WHAT IT ENCODES

Primary membranous is the most common cause of primary nephrotic syndrome in older/middle-aged white adults. In adults >60, screen for an underlying malignancy (solid tumors) as a secondary trigger.

BOARD TRAP

Membranous is #1 primary nephrotic in WHITE ADULTS; MCD is #1 in children, FSGS leads in African-ancestry/ESKD.

13. 1/3 remit · 1/3 relapse · 1/3 ESKD

WHAT YOU SEE

Bottom: 1/3 REMIT, 1/3 RELAPSE, 1/3 ESKD; interstitial fibrosis → prognosis

WHAT IT ENCODES

Classic 'rule of thirds' natural history: one-third spontaneous remission, one-third relapsing/persistent, one-third progress to ESKD. Tubulointerstitial fibrosis on biopsy is the best histologic predictor of progression.

BOARD TRAP

Interstitial fibrosis — not the glomerular deposit stage — best predicts long-term renal outcome.
